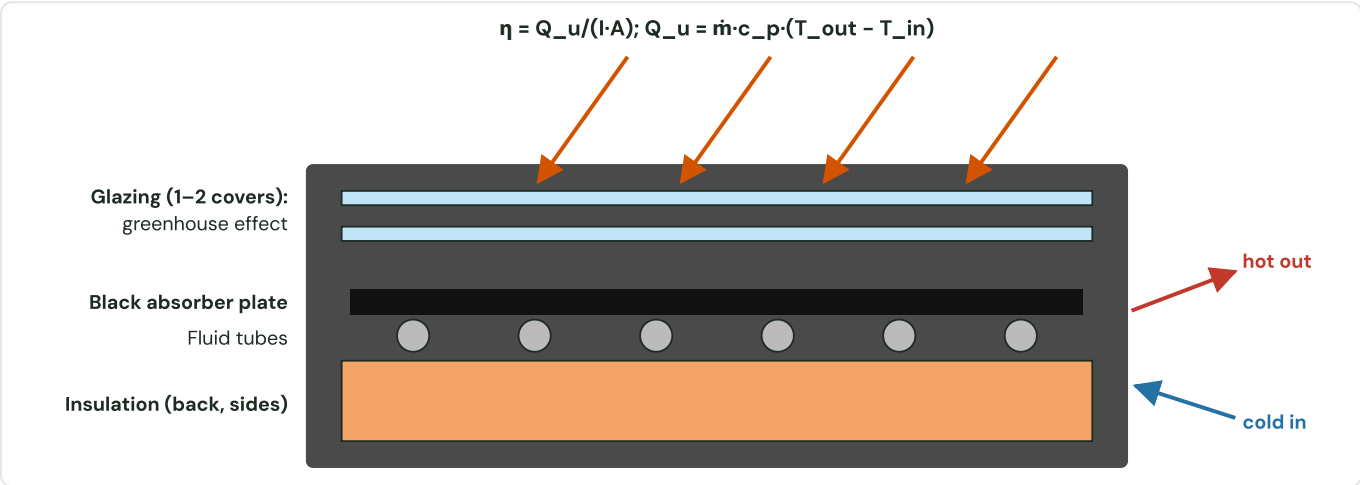
UNIT 7 · TOPIC 3

Solar Thermal Collection and Thermal Energy Storage

Syllabus topic: Solar energy collection for thermal and PV application and its storage

Flat-plate collector (FPC)



Flat-plate solar collector, non-concentrating, up to ~100 °C (redrawn from the source figure)

The workhorse low-temperature collector. Parts:

- One or two transparent **GLAZING** covers — trap long-wave re-radiation (the greenhouse effect) and cut convection loss.
- A black **ABSORBER** plate with bonded fluid **TUBES** (or an air passage).
- Back and side **INSULATION**, all in a weatherproof casing.

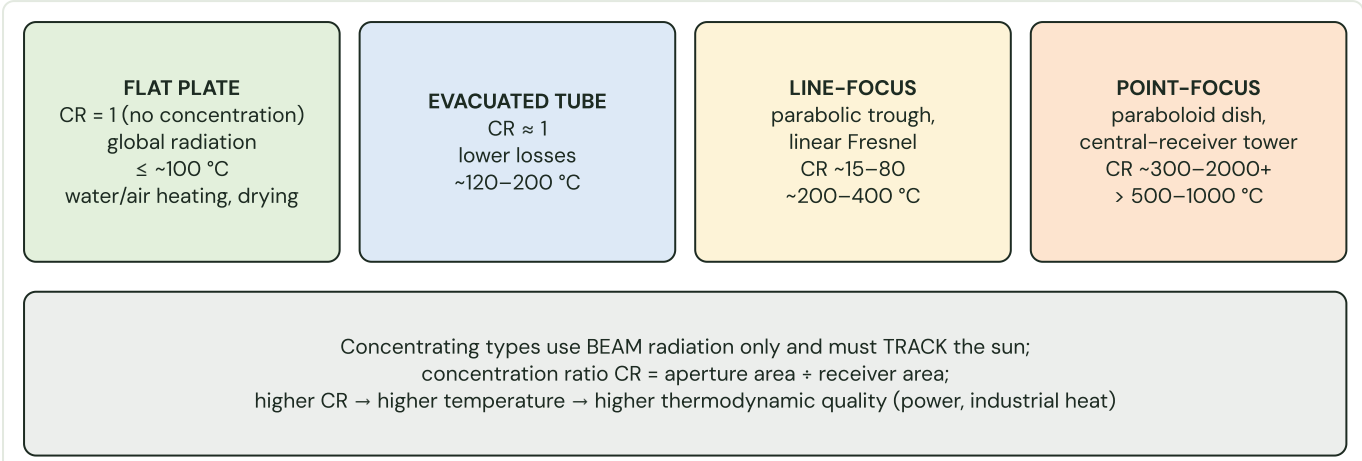
It collects BOTH beam and diffuse (GLOBAL) radiation, needs NO tracking and reaches up to **~100 °C** — for water heating, crop/grain DRYING (solar dryers) and space heating.

$$\text{Useful heat } Q_u = \dot{m} \cdot c_p \cdot (T_{out} - T_{in}); \text{ efficiency } \eta = Q_u/(I \cdot A)$$

$$\text{Hottel–Whillier–Bliss: } \eta = F_R(\tau\alpha) - F_R U_L (T_i - T_a)/I$$

- Efficiency FALLS as the plate runs hotter above ambient (losses ∝ ΔT).
- Selective absorber coatings (high absorptance α, low emittance ε) and evacuated-tube designs cut losses and raise the working temperature (**~120–200 °C**).

Concentrating collectors



Solar collectors by concentration and temperature (redrawn from the source figure)

$$\text{Concentration ratio } CR = \text{aperture area} \div \text{receiver area}$$

Type	Examples	CR	Temperature / use
LINE-FOCUS	Parabolic trough, linear Fresnel	~15–80	~200–400 °C — process heat, CSP power
POINT-FOCUS	Paraboloid DISH, central-receiver solar TOWER with heliostats	~300–2000+	> 500–1000 °C — power generation

- All concentrators use ONLY beam radiation and must TRACK the sun (single- or two-axis).
- Higher temperature = higher thermodynamic quality, but more optical/tracking complexity.

Thermal energy storage

Sunshine is intermittent, so storage bridges night and cloud.

Type	How	Examples
SENSIBLE heat	Raise the temperature of a medium: Q = m·c_p·ΔT	Water (high c_p 4.19 kJ/kg·K, cheap); rock/pebble beds (air systems); soil/concrete
LATENT heat	Phase-change materials store large heat at nearly constant temperature while melting (high energy density)	Salt hydrates, paraffins
THERMOCHEMICAL	Reversible reactions/sorption	Research/advanced

For crop DRYING, storage can be as simple as a rock bed or the grain mass itself; for cooking, insulated hot-box retention.

Practice MCQs

Q36. A flat-plate collector primarily collects:

(a) Beam only
(b) Global (beam + diffuse) radiation
(c) Diffuse only
(d) UV only

Q37. The glazing cover of a flat-plate collector works by:

(a) Reflecting all light
(b) The greenhouse effect — trapping long-wave re-radiation, cutting convection
(c) Cooling the plate
(d) Concentrating light

Q38. A flat-plate collector typically reaches temperatures up to about:

(a) 30 °C
(b) 100 °C
(c) 400 °C
(d) 1000 °C

Q39. Useful heat from a collector is Q =

(a) $mc_p\Delta T$ ($\dot{m}\cdot c_p\cdot(T_{out}-T_{in})$)
(b) I·A only
(c) mgh
(d) ½mv²

Q40. Collector efficiency as plate temperature rises above ambient:

(a) Rises
(b) Falls (losses ∝ ΔT)
(c) Constant
(d) Zero

Q41. A selective absorber surface has:

(a) Low absorptance
(b) High absorptance and low emittance
(c) High emittance
(d) Mirror finish

Q42. Concentrating collectors use which radiation and require:

(a) Diffuse, no tracking
(b) Beam radiation and sun tracking
(c) Global, fixed
(d) UV, tracking

Q43. Concentration ratio (CR) is:

(a) Receiver ÷ aperture area
(b) Aperture ÷ receiver area
(c) Beam ÷ diffuse
(d) Temperature ratio

Q44. A parabolic trough is a:

(a) Point-focus collector
(b) Line-focus collector (~200–400 °C)
(c) Flat plate
(d) Evacuated tube

Q45. The highest temperatures (>500–1000 °C) are reached by:

(a) Flat plate
(b) Evacuated tube
(c) Point-focus dish / central receiver
(d) Trough

Q46. Sensible-heat storage stores energy by:

(a) Melting a material
(b) Raising a medium's temperature ($Q = mc_p\Delta T$)
(c) A chemical reaction
(d) Compression

Q47. The cheap, common sensible-heat storage medium for solar water heating is:

(a) Rock
(b) Water (high c_p)
(c) Sand
(d) Oil

Q48. A phase-change material (PCM) stores:

(a) Sensible heat only
(b) Latent heat at nearly constant temperature
(c) Electricity
(d) Momentum

Q49. For solar AIR heating/drying, a suitable storage is a:

(a) Battery
(b) Rock/pebble bed
(c) Capacitor
(d) Flywheel

Answers

Q36 — (b) Global (beam + diffuse) radiation

Q37 — (b) The greenhouse effect — trapping long-wave re-radiation, cutting convection — *Glazing traps long-wave re-radiation (greenhouse effect).*

Q38 — (b) 100 °C
Q39 — (a) mc_pΔT (ṁ·c_p·(T_out-T_in))
Q40 — (b) Falls (losses ∝ ΔT) — *Efficiency falls as (T_i - T_a)/I rises — the Hottel–Whillier–Bliss line.*
Q41 — (b) High absorptance and low emittance
Q42 — (b) Beam radiation and sun tracking — *Concentrators use beam only and track.*
Q43 — (b) Aperture ÷ receiver area — *CR = aperture/receiver.*
Q44 — (b) Line-focus collector (~200–400 °C) — *Line-focus (trough/Fresnel) 200–400 °C.*
Q45 — (c) Point-focus dish / central receiver — *Point-focus (dish/tower) > 500 °C.*
Q46 — (b) Raising a medium's temperature (Q = mc_pΔT)
Q47 — (b) Water (high c_p)
Q48 — (b) Latent heat at nearly constant temperature
Q49 — (b) Rock/pebble bed

Sources: Sukhatme & Nayak — Solar Energy; Duffie & Beckman — Solar Engineering of Thermal Processes; Rai — Non-Conventional Energy Sources; NREL/ITRPV and manufacturer datasheets (efficiency figures as given in the source notes, July 2026); MNRE guidelines; NPTEL solar-energy lectures.